TEAM ACE PROJECT PRESENTATION

SMART DRAIN & HEATWAVE RESPONSE SYSTEM

An integrated urban climate management platform that uses real-time weather API forecasts and IoT sensor data to proactively address **heatwave droughts** (automated micro-irrigation) and **torrential rain** (smart pre-storm drainage) using a single rainwater storage tank.

PROJECT TITLE
Smart Drain & Heatwave System

TEAM
Team ACE

Extreme Summer Climate Hazards in Urban Environments

Modern cities face compound climate hazards where heatwaves and flash floods occur in rapid succession within short time windows.

🔥 1. Heatwave & Drought

- Extreme summer heat dries soil below 30% moisture.
- Urban heat island effect causes plant wilting and severe pedestrian discomfort.
- Manual watering wastes valuable municipal tap water.

🌧 2. Torrential Rain & Flooding

- Sudden downpours instantly exceed local storm sewer capacity.
- Rainwater tanks fill to 100% capacity and overflow into streets.
- Lack of pre-storm drainage leads to localized urban flooding.

⚠ 3. Limits of Conventional Systems

- Existing drains operate as one-way reactive systems.
- Stormwater is dumped into sewers instead of being stored.
- No integration between flood prevention and heatwave irrigation.

💡 The Need for an Integrated Solution

Urban infrastructure requires a smart system capable of automatically switching between **storing rainwater for drought irrigation** and **pre-draining tanks before severe storms land**.

One Tank, Two Jobs: Bi-Directional Climate Management

Team ACE's SDS Platform utilizes a single 6 L storage tank to manage both heatwave irrigation and pre-storm flood protection.

🌱 JOB 1: Heatwave Micro-Irrigation (P1 Pump)

- **Trigger Condition:** Clear weather (no rain forecast), Soil moisture < 30%, Temp ≥ 30°C, and Tank storage > 20%.
- **Control Action:** Activates **P1 Irrigation Pump** to deliver a 12-second pulse (~0.34 L) to the crop box.
- **Benefit:** Recycles stored rainwater to preserve urban crops without wasting tap water.

🌧 JOB 2: Pre-Storm Flood Drainage (P2 Pump)

- **Trigger Condition:** KMA Open API receives a Heavy Rain Forecast AND Tank storage level > 20%.
- **Control Action:** Activates **P2 Pre-drain Pump** to empty the tank down to 10% before downpour hits.
- **Benefit:** Creates a empty storage buffer in advance, preventing urban sewer overload.

🔄 The Rainwater Circularity Loop

The torrential downpour that threatens street flooding is stored into the tank, becoming the exact water source that cools the city during the next heatwave.

Dumb Edge Node, Smart Hub System Architecture

1. ESP32 Edge Node

- **Sensors:** Capacitive Soil Moisture (ADC1), DHT22 Temp/Humidity, HC-SR04 Ultrasonic Tank Level.
- **Actuators:** 2-Channel 5V Relay controlling P1 & P2 Submersible Pumps.
- **Role:** Reads telemetry every 2 s, reports via MQTT, executes commands with 15 s safety watchdog.

2. Raspberry Pi Main Hub

- **Mosquitto MQTT Broker:** Handles real-time JSON messaging.
- **decision.py:** Central decision engine evaluating state logic every 5 s.
- **FastAPI & SQLite:** Serves system dashboard and records event telemetry logs.

3. KMA Forecast API

- **Open API Polling:** Fetches short-term forecast every 10 min.
- **Grid Conversion:** Converts lat/lon to Lambert conformal conic grid.
- **Fail-Safe:** API outages default to "last valid forecast" then "no rain", preventing erroneous drainage.

Telemetry Payload (ESP32 → Pi, 2 s)

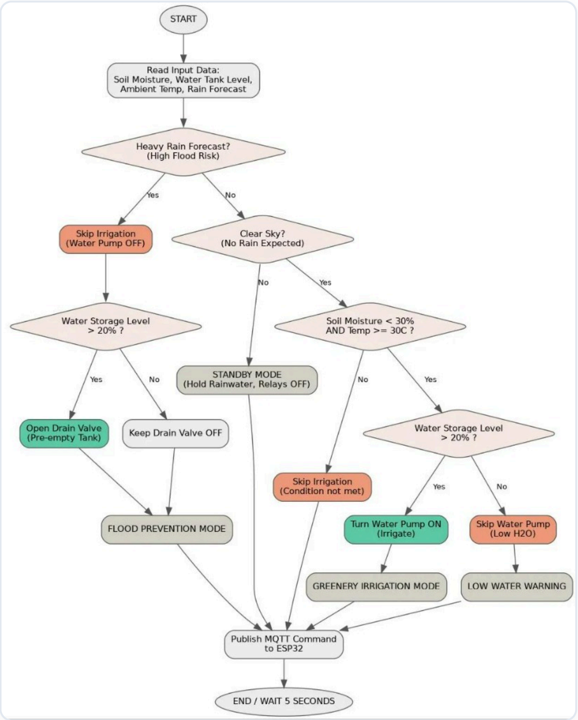
{ "ts": 1786342072.4, "node": "ace-esp32", "soil_moisture_pct": 27.4, "ambient_temp_c": 34.2, "tank_level_pct": 43.1 }

Command Payload (Pi → ESP32, 5 s)

{ "mode": "GREENERY_IRRIGATION", "pump": true, "drain": false, "reason": "moisture 27%, 34.2C", "ttl_s": 15.0 }

5-Second Real-Time Decision Algorithm Flowchart

[System Decision Algorithm Flowchart]



Six Operating Modes Summary

Operating Mode	Description & Action
FLOOD_PREVENTION	P2 Pre-drain active (empties tank to ≤ 10%)
GREENERY_IRRIGATION	P1 Water Pump 12 s pulse ON (~0.34 L micro-dose)
IRRIGATION_SOAK	Waits 10 min for soil absorption before next evaluation
LOW_WATER_WARNING	Hot & dry, but tank storage level ≤ 20% (holds water)
STANDBY	Normal monitoring mode (holds harvested rainwater)
FAIL_SAFE	Sensor/API outage — all relays shut OFF for safety

* **Strict Priority Rule:** Flood prevention always overrides irrigation. A rain forecast arriving mid-irrigation instantly cancels P1 and activates P2 pre-drainage.

Conventional Systems vs. Team ACE SDS Smart Platform

Feature	Conventional Municipal Drainage	ACE SDS Smart Platform
Primary Goal	Flood prevention only	**Flood + Heatwave + Drought integrated control**
Rainwater Processing	Store then discharge into sewer (wasted)	**Filtered ➔ Stored ➔ 100% Recycled for Irrigation & Mist**
Water Flow Mechanism	One-way gravity discharge	**Bi-directional smart routing (P1 Irrigation & P2 Pre-drain)**
Decision Timing	Reactive response after heavy rain	**Proactive 1-hour KMA forecast API poller**
Application Scope	Large public civil infrastructure	**Roofs, planter boxes, smart shade micro-zones**

🌫️ Smart Shade Cooling Fog Integration & Innovations

💧 Rainwater First
Prioritizes stored rainwater over municipal tap water.

🎯 Multi-Height Spray
Nozzle heights for children (1.2m) & adults (1.8m).

🌡️ IoT Climate Control
Lowers felt ambient temperature by 3~5°C.

🛡️ 3-Tier Filtration
Topsoil, fabric filter, and gravel prevent clogging.

1:1 Physical Prototype Live Demonstration

LIVE PROTOTYPE DEMONSTRATION

"Let's Watch the Physical Prototype Demonstration Video"

Demonstrating the 1:1 physical hardware prototype — featuring real-time water pump switching for P1 (Heatwave Irrigation) and P2 (Flood Pre-drainage) based on live forecast API signals and sensor telemetry.

🌱 **1. Heatwave Micro-Irrigation (P1 Pulse) Action**

2. Pre-Storm Smart Flood Drainage (P2) Action

07 — CONCLUSION & Q&A

SMART DRAIN & HEATWAVE RESPONSE SYSTEM

Thank you for your time. Ready for any questions.

Control Logic	Simulation Unit Tests	Weather API Grid
100% Verified	**15 / 15 PASS**	**KMA Forecast Validated**

Question & Answer (Q&A)

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